Pandas: Top 200 Functions and Methods

Function / method name, common syntax, and purpose | English reference

No.	Function / Method	Common Syntax	Purpose
1	Series	pd.Series()	Create a one-dimensional labeled array.
2	DataFrame	pd.DataFrame()	Create a two-dimensional labeled table.
3	Index	pd.Index()	Create an immutable sequence of axis labels.
4	RangeIndex	pd.RangeIndex()	Create an integer index with a regular range.
5	date_range	pd.date_range()	Generate a fixed-frequency sequence of dates.
6	period_range	pd.period_range()	Generate a sequence of time periods.
7	timedelta_range	pd.timedelta_range()	Generate a sequence of time differences.
8	read_csv	pd.read_csv()	Read a CSV file into a DataFrame.
9	read_excel	pd.read_excel()	Read an Excel file into a DataFrame.
10	read_json	pd.read_json()	Read a JSON file or JSON string.
11	read_parquet	pd.read_parquet()	Read a Parquet file.
12	read_sql	pd.read_sql()	Read a SQL query or table into a DataFrame.
13	read_sql_query	pd.read_sql_query()	Read the result of a SQL query.
14	read_sql_table	pd.read_sql_table()	Read a SQL database table.
15	read_html	pd.read_html()	Read HTML tables from a webpage or file.
16	read_xml	pd.read_xml()	Read XML data into a DataFrame.
17	read_fwf	pd.read_fwf()	Read a fixed-width formatted text file.
18	read_clipboard	pd.read_clipboard()	Read text from the system clipboard.
19	read_pickle	pd.read_pickle()	Load a pickled pandas object.
20	read_feather	pd.read_feather()	Read a Feather-format file.
21	to_csv	df.to_csv()	Export a DataFrame to a CSV file.
22	to_excel	df.to_excel()	Export a DataFrame to an Excel file.
23	to_json	df.to_json()	Convert a DataFrame to JSON.
24	to_parquet	df.to_parquet()	Export a DataFrame to Parquet format.
25	to_sql	df.to_sql()	Write a DataFrame to a SQL table.
26	to_html	df.to_html()	Render a DataFrame as an HTML table.
27	to_xml	df.to_xml()	Convert a DataFrame to XML.
28	to_dict	df.to_dict()	Convert a DataFrame to a dictionary.
29	to_records	df.to_records()	Convert a DataFrame to a record array.
30	to_numpy	df.to_numpy()	Return the data as a NumPy array.
31	head	df.head()	Return the first rows.
32	tail	df.tail()	Return the last rows.
33	sample	df.sample()	Return random rows or columns.
34	info	df.info()	Display concise information about the DataFrame.
35	describe	df.describe()	Generate descriptive statistics.
36	shape	df.shape	Return the number of rows and columns.
37	size	df.size	Return the total number of elements.
38	ndim	df.ndim	Return the number of dimensions.
39	axes	df.axes	Return the row and column axis labels.
40	dtypes	df.dtypes	Return the data type of each column.
41	select_dtypes	df.select_dtypes()	Select columns based on their data types.
42	memory_usage	df.memory_usage()	Estimate memory usage of each column.
43	empty	df.empty	Check whether the DataFrame has no elements.
44	keys	df.keys()	Return the column labels.

No.	Function / Method	Common Syntax	Purpose
45	columns	df.columns	Get or set column labels.
46	index	df.index	Get or set row labels.
47	get	df.get()	Retrieve a column or key with a default value.
48	loc	df.loc[]	Select rows and columns by labels.
49	iloc	df.iloc[]	Select rows and columns by integer positions.
50	at	df.at[]	Access one value by row and column labels.
51	iat	df.iat[]	Access one value by integer positions.
52	query	df.query()	Filter rows using a query expression.
53	filter	df.filter()	Select rows or columns by labels.
54	where	df.where()	Keep values where a condition is true.
55	mask	df.mask()	Replace values where a condition is true.
56	isin	df.isin()	Check whether values belong to given items.
57	between	Series.between()	Check whether values fall within an interval.
58	nlargest	df.nlargest()	Return rows with the largest values.
59	nsmallest	df.nsmallest()	Return rows with the smallest values.
60	duplicated	df.duplicated()	Mark duplicate rows.
61	drop_duplicates	df.drop_duplicates()	Remove duplicate rows.
62	drop	df.drop()	Remove specified rows or columns.
63	pop	df.pop()	Remove and return a column.
64	insert	df.insert()	Insert a column at a chosen position.
65	assign	df.assign()	Add columns or return a modified DataFrame.
66	rename	df.rename()	Rename index labels or columns.
67	rename_axis	df.rename_axis()	Set the names of index or column axes.
68	set_index	df.set_index()	Set one or more columns as the index.
69	reset_index	df.reset_index()	Reset the index to default integer labels.
70	reindex	df.reindex()	Conform data to a new index.
71	reindex_like	df.reindex_like()	Match another object's index and columns.
72	set_axis	df.set_axis()	Assign new labels to an axis.
73	sort_values	df.sort_values()	Sort rows by column values.
74	sort_index	df.sort_index()	Sort by index labels.
75	rank	df.rank()	Compute numerical ranks of values.
76	round	df.round()	Round numeric values to a chosen precision.
77	abs	df.abs()	Return absolute numeric values.
78	add	df.add()	Add values with label alignment.
79	sub	df.sub()	Subtract values with label alignment.
80	mul	df.mul()	Multiply values with label alignment.
81	div	df.div()	Divide values with label alignment.
82	floordiv	df.floordiv()	Perform floor division.
83	mod	df.mod()	Calculate remainders.
84	pow	df.pow()	Raise values to powers.
85	dot	df.dot()	Calculate matrix multiplication.
86	add_prefix	df.add_prefix()	Add a prefix to column labels.
87	add_suffix	df.add_suffix()	Add a suffix to column labels.
88	combine	df.combine()	Combine two DataFrames using a function.
89	combine_first	df.combine_first()	Fill missing values using another DataFrame.
90	update	df.update()	Modify values using non-NA values from another object.

No.	Function / Method	Common Syntax	Purpose
91	compare	df.compare()	Show differences between two DataFrames.
92	equals	df.equals()	Test whether two DataFrames are equal.
93	all	df.all()	Check whether all values are true along an axis.
94	any	df.any()	Check whether any value is true along an axis.
95	sum	df.sum()	Calculate the sum.
96	prod	df.prod()	Calculate the product.
97	mean	df.mean()	Calculate the arithmetic mean.
98	median	df.median()	Calculate the median.
99	mode	df.mode()	Find the most frequent value.
100	min	df.min()	Find the minimum value.
101	max	df.max()	Find the maximum value.
102	count	df.count()	Count non-missing values.
103	std	df.std()	Calculate standard deviation.
104	var	df.var()	Calculate variance.
105	sem	df.sem()	Calculate standard error of the mean.
106	skew	df.skew()	Calculate skewness.
107	kurt	df.kurt()	Calculate kurtosis.
108	quantile	df.quantile()	Calculate requested quantiles.
109	cumsum	df.cumsum()	Calculate cumulative sums.
110	cumprod	df.cumprod()	Calculate cumulative products.
111	cummin	df.cummin()	Calculate cumulative minimum values.
112	cummax	df.cummax()	Calculate cumulative maximum values.
113	diff	df.diff()	Calculate differences between adjacent values.
114	pct_change	df.pct_change()	Calculate percentage change between rows.
115	cov	df.cov()	Calculate pairwise covariance.
116	corr	df.corr()	Calculate pairwise correlation.
117	corrwith	df.corrwith()	Calculate correlation with another object.
118	groupby	df.groupby()	Group data for aggregation or transformation.
119	agg	df.agg()	Apply one or more aggregation operations.
120	aggregate	df.aggregate()	Alias for agg; apply aggregation functions.
121	transform	df.transform()	Apply a function while preserving group shape.
122	apply	df.apply()	Apply a function along an axis.
123	applymap	df.map()	Apply a function elementwise to a DataFrame.
124	map	Series.map()	Map values using a function or mapping.
125	pipe	df.pipe()	Pass an object through a function.
126	rolling	df.rolling()	Create a rolling window calculation.
127	expanding	df.expanding()	Create an expanding window calculation.
128	ewm	df.ewm()	Create an exponentially weighted window.
129	resample	df.resample()	Group time-series data by a new frequency.
130	shift	df.shift()	Move data by a number of periods.
131	asfreq	df.asfreq()	Convert time-series data to a specified frequency.
132	asof	df.asof()	Return the last non-missing row before a date or label.
133	interpolate	df.interpolate()	Fill missing values using interpolation.
134	ffill	df.ffill()	Forward-fill missing values.
135	bfill	df.bfill()	Back-fill missing values.

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136	fillna	df.fillna()	Fill missing values with a chosen value or method.
137	dropna	df.dropna()	Remove missing values.
138	isna	df.isna()	Detect missing values.
139	notna	df.notna()	Detect non-missing values.
140	isnull	df.isnull()	Alias for isna; detect missing values.
141	notnull	df.notnull()	Alias for notna; detect non-missing values.
142	replace	df.replace()	Replace specified values.
143	infer_objects	df.infer_objects()	Infer better data types for object columns.
144	convert_dtypes	df.convert_dtypes()	Convert columns to suitable nullable data types.
145	astype	df.astype()	Cast data to specified data types.
146	to_numeric	pd.to_numeric()	Convert values to numeric types.
147	to_datetime	pd.to_datetime()	Convert values to datetime objects.
148	to_timedelta	pd.to_timedelta()	Convert values to timedelta objects.
149	cut	pd.cut()	Bin continuous values into intervals.
150	qcut	pd.qcut()	Bin values into quantile-based intervals.
151	factorize	pd.factorize()	Encode values as integer labels.
152	get_dummies	pd.get_dummies()	Create indicator variables from categorical data.
153	crosstab	pd.crosstab()	Compute a frequency table for categories.
154	pivot	df.pivot()	Reshape data using unique index-column pairs.
155	pivot_table	df.pivot_table()	Create a spreadsheet-style pivot table.
156	melt	pd.melt()	Unpivot columns from wide format to long format.
157	stack	df.stack()	Move a column level into the row index.
158	unstack	df.unstack()	Move an index level into columns.
159	wide_to_long	pd.wide_to_long()	Reshape wide data into long format.
160	explode	df.explode()	Turn list-like values into separate rows.
161	transpose	df.transpose()	Swap rows and columns.
162	T	df.T	Transpose rows and columns.
163	merge	pd.merge()	Join DataFrames using database-style keys.
164	join	df.join()	Join columns using index or key relationships.
165	concat	pd.concat()	Combine objects along rows or columns.
166	merge_ordered	pd.merge_ordered()	Merge ordered data with optional filling.
167	merge_asof	pd.merge_asof()	Merge by nearest key rather than exact equality.
168	append	—	Legacy DataFrame method removed; use pd.concat().
169	Series.str	s.str	Access string methods on Series values.
170	Series.dt	s.dt	Access datetime properties and methods.
171	Series.cat	s.cat	Access categorical properties and methods.
172	str.lower	s.str.lower()	Convert strings to lowercase.
173	str.upper	s.str.upper()	Convert strings to uppercase.
174	str.title	s.str.title()	Convert strings to title case.
175	str.strip	s.str.strip()	Remove leading and trailing characters.
176	str.replace	s.str.replace()	Replace matching text in strings.
177	str.contains	s.str.contains()	Check whether strings contain a pattern.
178	str.startswith	s.str.startswith()	Check whether strings start with a prefix.
179	str.endswith	s.str.endswith()	Check whether strings end with a suffix.
180	str.split	s.str.split()	Split strings around a separator.

No.	Function / Method	Common Syntax	Purpose
181	str.extract	s.str.extract()	Extract groups matched by a regular expression.
182	str.len	s.str.len()	Return the length of each string.
183	str.slice	s.str.slice()	Slice each string by position.
184	dt.year	s.dt.year	Get the year component from dates.
185	dt.month	s.dt.month	Get the month component from dates.
186	dt.day	s.dt.day	Get the day component from dates.
187	dt.dayofweek	s.dt.dayofweek	Get the weekday number.
188	dt.strftime	s.dt.strftime()	Format datetime values as strings.
189	dt.floor	s.dt.floor()	Round datetimes down to a frequency.
190	dt.ceil	s.dt.ceil()	Round datetimes up to a frequency.
191	dt.normalize	s.dt.normalize()	Set time components to midnight.
192	Categorical	pd.Categorical()	Create categorical data with defined categories.
193	unique	s.unique()	Return unique values in a Series.
194	nunique	s.nunique()	Count distinct values.
195	value_counts	s.value_counts()	Count occurrences of each distinct value.
196	to_string	df.to_string()	Render the DataFrame as plain text.
197	to_markdown	df.to_markdown()	Render the DataFrame as a Markdown table.
198	to_latex	df.to_latex()	Render the DataFrame as LaTeX tabular text.
199	to_clipboard	df.to_clipboard()	Copy the DataFrame to the clipboard.
200	copy	df.copy()	Create a copy of the object.